Collapsing the dispatch ceiling: kernel fusion for browser-native WebGPU quantum-circuit simulation

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Abstract

Sequential GPU workloads in the browser are throttled by a fixed per-dispatch cost: every WebGPU compute dispatch pays a driver/submission overhead that, for fine-grained work, dominates the actual arithmetic. We quantify this overhead for browser quantum-circuit simulation — a per-gate dispatch cost of $\alpha \approx 5$–$500\,\mu$s on commodity hardware — and show that **kernel fusion**, executing a chain of k quantum gates inside a single WGSL `main()`, drives the effective per-gate cost toward $\alpha/k + C$. We implement fusion as hand-written dense-tile WGSL kernels (2-, 3-, and 4-qubit dense operators) and measure a tier ladder against the simulator's own unfused multi-dispatch path: brick-wall two-qubit fusion reaches **3.04**×, three-qubit-tile (Tier-C) fusion **4.22**×, both at fidelity $F \geq 0.99999$ vs the unfused statevector. Four-qubit-tile (Tier-D) fusion is an **honest negative** — it tops out at 3.78× because it crosses from the dispatch-bound into the compute-bound regime, where the per-block complex-multiply count grows faster than the memory traffic fusion saves. Fusion is correctness-preserving — across 360 (N, k, trial) configurations the worst fidelity loss against the unfused statevector is $1 - F = 2.4 \times 10^{-6}$ — and pushes the fused statevector update toward memory-bandwidth-bound throughput. All results run in a browser tab with no installation; every number is backed by a committed, environment-stamped JSON artifact. We do not benchmark against native GPU simulators (cuQuantum [1], Qiskit Aer [2]) — those require hardware unavailable to us — and position relative to them only through the bandwidth-fraction argument.

1 Introduction

A WebGPU compute dispatch is not free. Beyond the shader's own runtime, each `dispatchWorkgroups` call pays a driver-side submission and scheduling cost. For coarse-grained work this is negligible; for *fine-grained, sequential* work — where the natural unit of computation is small and many units run in strict order — it becomes the dominant term. Quantum-circuit statevector simulation is a clean instance: each gate is a small (2×2, 4×4) operator applied to a large amplitude array, and gates within a circuit must be applied in order. A one-gate-per-dispatch scheduler spends more time in the driver than in the ALUs.

We measure this overhead directly, establish the per-gate cost as a function of circuit width and fusion factor, and show that fusing gate chains into single dispatches collapses it. The contribution is not a new simulation algorithm — statevector simulation is textbook — but a quantification of the WebGPU dispatch ceiling for sequential workloads and a fidelity-validated demonstration of how far kernel fusion moves it, including the regime where fusion stops paying off. Single-kernel fusion has previously removed this overhead for two other sequential WebGPU workloads — evolutionary fitness evaluation, at 159–720× [3], and autoregressive transformer decoding, at 66–458× [4] — both of which are deeply dispatch-bound. Quantum-circuit statevector simulation is a third domain, and, as we show, one that sits far closer to the bandwidth-bound end of that spectrum:

each gate already moves the entire 2^N-amplitude statevector, so the per-dispatch overhead is a much smaller fraction of the work than in those deeply dispatch-bound workloads, and dense-tile fusion additionally crosses into a compute-bound regime (§3.3). The same technique that delivers hundreds-fold speedups there therefore caps at a few-fold here. Locating where a workload sits on that dispatch-bound-to-bandwidth-bound spectrum is the point. The work runs entirely in a browser tab.

2 Methods

2.1 Simulator and fusion kernels

Amplitudes are stored as interleaved `vec2<f32>` (re, im) in a storage buffer of size 2^{N+3} bytes for N qubits. A single-qubit gate dispatches $N/2$ threads, each updating the amplitude pair that differs in the target bit; a controlled two-qubit gate dispatches $N/4$ threads. Fusion replaces a chain of such gates with one dense operator applied in a single dispatch: a brick-wall layer of adjacent two-qubit gates becomes one dense 4×4 dispatch (Tier-B); a three-qubit tile becomes a dense 8×8 dispatch (Tier-C); a four-qubit tile a dense 16×16 dispatch (Tier-D). The dense-tile operators are hand-written WGSL (`two-qubit-dense.wgsl`, `three-qubit-dense.wgsl`, `four-qubit-dense.wgsl`); the fused operator matrix is precomputed on the host and uploaded once per fused block. (This is fixed-tile fusion, not a general gate-chain JIT compiler.)

2.2 Measurement harness

Wall-clock is measured with `performance.now()` bracketed by a forced GPU sync (a mapped readback of a tiny buffer) before and after, because `queue.submit` is non-blocking and raw timing is otherwise fiction. Five warmup samples are discarded and twenty retained; we report the median and flag any configuration with std/median > 0.1 as `noisy`. Correctness uses fidelity $F = |\langle\psi_{\text{ref}}|\psi_{\text{test}}\rangle|^2$ against the unfused statevector, with a pass bar of $F \geq 1 - 10^{-5}$ for f32 amplitude paths. Every run emits a JSON artifact recording git SHA, `adapter.info`, WebGPU limits, OS, seeds, warmup/trials, and the pass bar.

3 Results

All measurements on an M2 Pro under Chromium with WebGPU.

3.1 The dispatch ceiling and its collapse

The per-gate dispatch cost is $\alpha \approx 5$–$500\,\mu$s on commodity WebGPU (level-1 dispatch-overhead experiment). Fusing k gates per dispatch drives the effective per-gate cost as $\alpha_{\text{eff}}(k) = \alpha/k + C$: measured single-qubit chains at $N = 8$ fall from $54.5\,\mu$s/gate at $k = 1$ to $17.8\,\mu$s at $k = 4$ and $\approx 15.8\,\mu$s at $k = 8$, the $1/k$ signature of amortizing a fixed cost over k operations. (This experiment is flagged `noisy` — sub-$20\,\mu$s timings sit near the harness's sync-fence resolution — but the $1/k$ trend is unambiguous across widths.)

3.2 Correctness

Fusion is correctness-preserving. Across 360 (N, k, trial) cells the worst fidelity is $F = 0.99999762$ $(1 - F = 2.4 \times 10^{-6})$, inside the 10^{-5} bar; the worst state-norm deviation is 10^{-6}. Fusion changes *when* arithmetic happens, not *what* it computes.

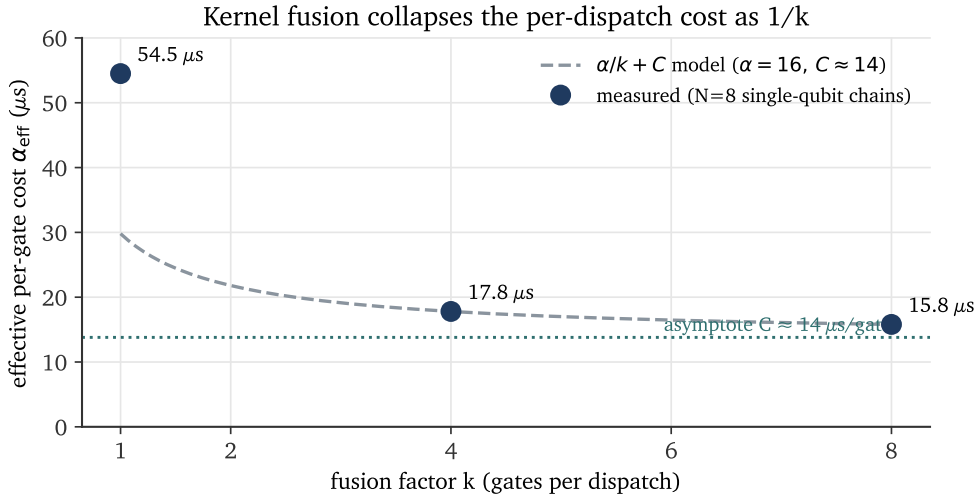

Figure 1: The per-gate dispatch cost collapses as $1/k$. Measured effective per-gate cost for N=8 single-qubit chains (markers) against the $\alpha/k + C$ model (dashed). The cost falls toward a fixed asymptote $C \approx 14\,\mu$s/gate as the fusion factor k grows; the k=1 point sits above the asymptotic model because the first unfused dispatch carries additional fixed submission overhead.

3.3 The fusion tier ladder

tier	fusion	bound	best speedup	worst F	verdict
B (brick-wall)	2-qubit → dense 4×4	dispatch	**3.04**×	1.0000000	pass ($\geq 2\times$)
C (three-qubit tile)	5 ops → dense 8×8	dispatch	**4.22**×	0.9999988	pass ($\geq 3\times$)
D (four-qubit tile)	7 ops → dense 16×16	dispatch→compute	3.78×	0.9999995	**fail ($\geq 5\times$)**

Best speedups measured at: B — N=20, depth 80; C — N=15, depth 80; D — N=20, depth 80.

Tier-D is the informative result. Its correctness is fine (worst $F = 0.9999995$), but it tops out at 3.78× against a 5× target because the four-qubit dense block performs 256 complex multiplies per 16-amplitude tile (16 per amplitude) — a 4× growth in per-tile arithmetic over Tier-C's 64, against only a 2× growth in the memory traffic that fusion eliminates. Tier-D crosses out of the dispatch-bound regime into the compute-bound one, where collapsing dispatches no longer helps. This locates the boundary of the technique rather than hiding it.

3.4 Toward bandwidth-bound throughput

In the dispatch-bound regime (small N, long circuits) fused same-qubit chains at $k = 32$ reach 2.62× ($N = 14$, depth 160). It is tempting to read the fused path's *logical* effective throughput — the statevector traffic of all D gates, $16\,D\,2^N/t$, which rises to 866 GB/s at $N = 20$, depth 160 — as memory-bandwidth saturation. It is not: that figure exceeds this device's ≈200 GB/s DRAM bandwidth by 4.3× precisely *because* fusion is working — a $k = 32$ chain reads each amplitude from DRAM once, applies all 32 gates in registers, and writes once, so it does the work of 866 GB/s of naive per-gate traffic while moving only one read+write pass per fused dispatch. The *physical* DRAM bandwidth, $16\lceil D/k \rceil 2^N/t$, peaks at just ∼27 GB/s — **13.5% of the device bound** — so the fused kernel is *not* bandwidth-saturated in this regime. The tell is in the artifact: the fused

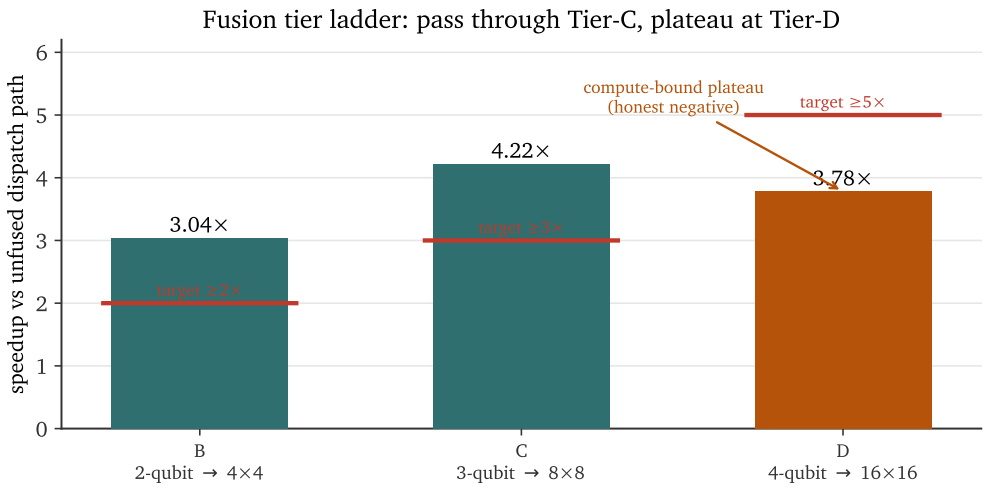

Figure 2: Fusion tier ladder. Brick-wall (B) and three-qubit-tile (C) fusion clear their speedup targets; four-qubit-tile (D) fusion plateaus at 3.78× against a 5× target because the dense 16×16 block is compute-bound — 256 complex multiplies per 16-amplitude tile (16 per amplitude), a 4× growth in per-tile arithmetic over Tier-C's 8×8 block (64 per tile), while the memory traffic fusion eliminates grows only 2×. The plateau is committed as a `status:"fail"` artifact, not hidden.

wall-time is pinned at ∼3 ms across $N = 10$ to $N = 20$ (a 1000× range of state sizes), i.e. floor-limited by the harness sync fence rather than by memory traffic. The fused-vs-unfused speedup compresses toward 1.25× at low depth ($N = 20$, depth 40) because few dispatches remain to remove, not because bandwidth is saturated; at longer depth the dispatch term still dominates and the gain persists ($N = 20$, depth 160 stays ∼2.6×). True DRAM saturation is the asymptotic ceiling fusion drives *toward* as N grows past the floor regime; demonstrating it empirically — at statevector sizes where the per-pass compute time dwarfs the ∼3 ms fence — is future work (§4). All figures here are derived from the committed `E10-throughput` artifact's (N, D, k, t) fields.

4 Honest limitations

- **Tier-D plateau (§3.3)** — fusion past three-qubit tiles is compute-bound and does not reach the 5× target. Committed as a `status:"fail"` artifact with the compute-vs-memory diagnosis.
- **Bandwidth saturation is asserted asymptotically, not measured.** At the widths we test ($N \leq 20$) the fused wall-time is floor-limited by the harness sync fence (∼3 ms, constant across a 1000× range of state sizes), so physical DRAM utilization peaks at only ∼13.5% of the device bound. We therefore do *not* claim the fused kernel saturates memory bandwidth in the measured regime; establishing the true asymptotic ceiling needs statevector sizes large enough that per-pass compute dwarfs the fence ($N \gtrsim 24$), a measurement we have not yet run.
- **No native-simulator head-to-head.** We measure fusion against this simulator's own unfused dispatch path. We did *not* benchmark cuQuantum custatevec [1] or Qiskit Aer GPU [2] — both require NVIDIA hardware we do not have — and we make no head-to-head claim

against them. Positioning is via the bandwidth-fraction argument only.

- **Fixed-tile fusion, not a JIT chain compiler.** The dense operators are hand-written for 2/3/4-qubit tiles; a general gate-chain fuser that emits WGSL per circuit is future work.
- **f32 amplitudes** — the 10^{-5} fidelity bar reflects single-precision GPU arithmetic; f64 is not available in WebGPU 1.0 [5].
- **No subgroup operations** — WebGPU subgroups (shuffle/reduce) are out of the 1.0 spec [5]; they would unlock a further $\sim 2\times$ on the reduction-heavy paths when available.

5 Related work

Per-dispatch overhead is the dominant cost for fine-grained sequential GPU work, which is exactly what fusion targets. Native GPU statevector simulators — NVIDIA cuQuantum custatevec [1] and Qiskit Aer GPU [2] — are the performance references for the algorithm, and both are known to approach memory-bandwidth-bound throughput for long contiguous gate chains; we target the same bandwidth-fraction regime but in a browser, and do not benchmark against them directly (§4). Kernel/operator fusion to amortize launch overhead is standard in GPU machine-learning runtimes; the contribution here is its quantification and fidelity-validated application to ordered quantum-gate chains under the specific constraints of WebGPU [5] in a browser tab. This is the third domain in which we characterize single-kernel WebGPU fusion: prior work established the framework for evolutionary fitness evaluation [3] and autoregressive transformer decoding [4], both deeply dispatch-bound and so reaching 10^2–$10^3\times$ speedups. The present domain is the instructive opposite end of the spectrum — each statevector gate already moves the full amplitude array, so the workload is far less dispatch-bound to begin with, and dense-tile fusion crosses into compute-bound at Tier-D (§3.3) — so fusion's payoff is bounded at a few-fold, and that plateau marks the crossover explicitly.

6 Software availability and reproducibility

Source: `github.com/abgnydn/webgpu-q` (MIT). Fusion kernels in `src/shaders/` (`*-dense.wgsl`); experiments in `experiments/level-3-fusion/` (E8 correctness, E9 dispatch-collapse, E10 throughput, E11 Tier-B, E12 Tier-C, E13 Tier-D); the committed `experiments/results/<date>/level-3/` JSON artifacts are the source of every number here. No installation: the simulator and its benchmarks run in any WebGPU-capable Chromium. Each artifact records git SHA, `adapter.info`, WebGPU device limits, OS, seeds, and the pass bar; timing is sync-fenced; warmup samples are discarded; failing configurations are committed with a diagnosis rather than rerun.

Generative-AI disclosure. Portions of the software, documentation, and this manuscript were drafted with a large language model used as a coding and writing aid; all output was author-reviewed, correctness was enforced by the fidelity tests and committed artifacts, and every quantitative claim is traceable to a recorded experiment.

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7 Conclusion

WebGPU's per-dispatch overhead caps fine-grained sequential GPU work in the browser. For quantum-circuit simulation it is $\alpha \approx 5$–$500\,\mu$s/gate, and kernel fusion drives the effective cost as $\alpha/k + C$: a tier ladder of dense-tile WGSL kernels delivers up to $4.22\times$ over the unfused path at fidelity ≥ 0.99999, with a clean, documented plateau at four-qubit tiles where the workload turns compute-bound. The result is a fidelity-validated map of how far fusion moves the dispatch ceiling — and exactly where it stops — entirely inside a browser tab, with every number reproducible from a URL.

References

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